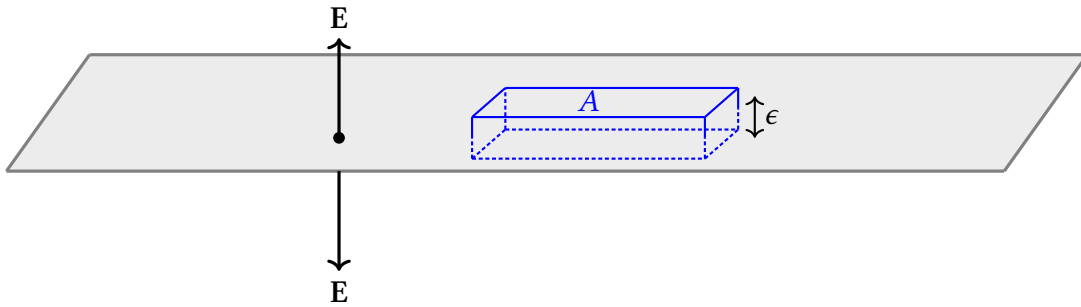


The Electric Field and Charge Distribution of an Infinite Charged Surface

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1 Infinite Plane

We draw a Gaussian pillbox on the surface, as shown below



The pillbox has length L_1 , width L_2 and infinitesimal height ϵ , where $L_1 L_2 = A$ and $L_1, L_2 \gg \epsilon \rightarrow 0$. We should consider two conditions,

1. The charge volume density of this pillbox ρ approaches the surface charge σ since

$$\rho = \frac{q_i}{A\epsilon} \Rightarrow q_i = \lim_{\epsilon \rightarrow 0} A\epsilon \cdot \frac{q_i}{A\epsilon} = A \lim_{\epsilon \rightarrow 0} \epsilon \cdot \frac{q_i}{A\epsilon} = A \times \frac{q_i}{A} = A\sigma$$

where q_i is the total charge of the pillbox. It shows that $\rho(\mathbf{r}) \rightarrow \sigma(\mathbf{r})$ so the Gauss's law gives that

$$\nabla \cdot \mathbf{E} = \frac{\sigma}{\epsilon_0} \quad (1)$$

2. The surface area of the pillbox $\mathcal{S}^2$ is given by

$$\mathcal{S}^2 = 2L_1 L_2 + 2\epsilon(L_1 + L_2) = 2A \Big|_{\epsilon \rightarrow 0}$$

By these two conditions, we can apply the convergence theorem on (1), and we obtain

$$\int_{\mathcal{V}} \nabla \cdot \mathbf{E} d\tau = \oint_{\mathcal{S}^2} \mathbf{E} \cdot d\mathbf{a} = \begin{cases} |\mathbf{E}| \cdot 2A \\ \frac{q_i}{\epsilon_0} = \frac{\sigma A}{\epsilon_0} \end{cases} \quad (2)$$

By (2), we have

$$\mathbf{E} = \frac{\sigma}{2\epsilon_0} \hat{\mathbf{n}} \quad (3)$$

Note that **the electric field $\mathbf{E}$ points away from the surface**. In addition, two points should be noted here. First, on either side of the charged plane, the magnitude of the electric field is $\sigma/2\epsilon_0$, however, a discontinuity arises at the interface, which will be discussed in the next section. Second, the magnitude of the electric field is independent of the distance from the plane. This is because, although the contribution from each individual charge decreases with distance, an increasing number of charges contribute as one moves farther away, and the combined effect remains the same. This is precisely why the plane is treated as infinite.

2 Boundary Conditions and Discontinuous

Suppose that we have a signed distance function $s(\mathbf{x})$ and a surface Ω , which satisfies

$$s(\mathbf{x}) = \begin{cases} +d(\mathbf{x}, \Omega) & \mathbf{x} \in \Omega^+ \\ -d(\mathbf{x}, \Omega) & \mathbf{x} \in \Omega^- \end{cases}$$

and

$$\Omega = \{\mathbf{x} \mid s(\mathbf{x}) = 0\}$$

We define $\nabla s = \hat{\mathbf{n}}$. The discontinuity(or jump) of the electric fields on the interface is defined as

$$\text{Jump} = \lim_{\mathbf{x} \rightarrow \Omega^+|_\epsilon} \mathbf{E}(\mathbf{x}) - \lim_{\mathbf{x} \rightarrow \Omega^-|_\epsilon} \mathbf{E}(\mathbf{x}) := \mathbf{E}^+ - \mathbf{E}^-$$

So that the discontinuity of the fields can be written as

$$\mathbf{E}^+ - \mathbf{E}^- = \frac{\sigma}{\epsilon_0} \hat{\mathbf{n}} \quad (4)$$

By the equation $\mathbf{E} = -\nabla V$, we obtain

$$\nabla V^+ - \nabla V^- = -\frac{\sigma}{\epsilon_0} \hat{\mathbf{n}} \quad (5)$$

Where we define $V^\pm \in \Omega^\pm|_\epsilon$. Eq (4) and (5) did show the discontinuity due to the different field directions. Note that the component of $\mathbf{E}$, which is parallel to the surface($E^\parallel$), doesn't contribute to the jump since

$$\sum \oint_{\text{one side}} \mathbf{E} \cdot d\ell = \sum_i \left(E^\parallel{}^+ L_i - E^\parallel{}^- L_i \right) = 0$$

as shown in the figure above. Furthermore, we define

$$\nabla V \cdot \hat{\mathbf{n}} := \frac{\partial V}{\partial \hat{\mathbf{n}}} = \partial_{\hat{\mathbf{n}}} V \quad (6)$$

By Eq (5) and Eq (6), we have

$$\boxed{\partial_{\mathbf{n}} V^+ - \partial_{\mathbf{n}} V^- = -\frac{\sigma}{\epsilon_0}} \quad (7)$$

This is called **normal derivative** of V . However, the potential is continuous across the interface in the sense of traces; on the other hand, the potential is uniform on the surface.

1. **Continuous Potential:** We define an arbitrary reference point $\mathcal{O} \in \Omega^+$, and $\forall \mathbf{x} \in \Omega^+$, we define the potential

$$V(\mathbf{x}) - V(\mathcal{O}) = - \int_{\mathcal{P}(\mathcal{O} \rightarrow \mathbf{x})} \mathbf{E}(\mathbf{x}') \cdot d\ell' \quad (8)$$

where $\mathcal{P}$ denotes the smooth path between the two chosen points. Consider two points $\mathbf{x}, \mathbf{x}' \in \Omega^+$ and a path $\mathcal{P}(\mathbf{x}' \rightarrow \mathbf{x})$ with length $\|\mathcal{P}\|$, where $\mathbf{x}' \rightarrow \mathbf{x}$. By Eq (8), we have

$$|V(\mathbf{x}) - V(\mathbf{x}')| \leq - \int |\mathbf{E}| d\ell \leq \left(\sup_{\mathcal{P} \in \Omega^+} |\mathbf{E}| \right) \|\mathcal{P}\|_{\mathbf{x}' \rightarrow \mathbf{x}} \quad (9)$$

While $\|\mathcal{P}\| \rightarrow 0$, so 9 gives that

$$\lim_{\mathbf{x}' \rightarrow \mathbf{x}} |V(\mathbf{x}) - V(\mathbf{x}')| = 0 \quad (10)$$

Therefore, we can claim that V is **continuous on $\Omega^\pm$** .

2. **Uniform Potential:** Consider the potential $V^+ \in \Omega^+|_\epsilon$ and $V^- \in \Omega^-|_\epsilon$, by the definition of the potential (8), we have

$$V^+ - V^- = - \int_{\mathbf{x}-\epsilon}^{\mathbf{x}+\epsilon} \mathbf{E} \cdot d\ell \rightarrow 0$$

where $\mathbf{x} \in \Omega$ and $|\epsilon| \rightarrow 0$. Then we can conclude that

$$V^+ = V^- \quad (11)$$

These properties naturally lead to several questions. Although the discontinuities that appear at interfaces may at first seem problematic or even inapplicable in certain cases, they in fact represent the most natural configuration. Such behavior is entirely dictated by the underlying charge distribution. A detailed discussion will be given in the next section. When the interface is taken to be the boundary surface of a conductor, these properties allow us to derive several distinctive features of conductors.

3 Surface Charge Distribution and the Force on a Conductor

Before discussing the properties of a conductor surface, we should verify

proposition 3.1. *The expression*

$$\varrho(\mathbf{x}) = \sigma(\mathbf{x})\delta(s(\mathbf{x}))$$

defines a distribution whose action on test functions reproduces the surface charge density σ on Ω .

In fact, we can check this property with the test function and the delta function $f(\mathbf{x})$, that is

$$\begin{aligned} \int_{\mathbb{R}^3} \varrho(\mathbf{x}) f(\mathbf{x}) d^3\mathbf{x} &= \int_{\mathbb{R}^3} \sigma(\mathbf{x})\delta(s(\mathbf{x})) f(\mathbf{x}) d^3\mathbf{x} \\ &= \int_{\Omega} \frac{\sigma(\mathbf{x})f(\mathbf{x})}{|\nabla s|} d^2\mathbf{x} = \int_{\Omega} \sigma(\mathbf{x})f(\mathbf{x}) d^2\mathbf{x} \end{aligned}$$

This indicates that the surface charge density σ is a regular function defined on the surface, while the delta function $\delta(s)$ is used solely to embed the surface distribution into three-dimensional space.

We rewrite the potential into the distribution form with the step function $H(x)$, which leads to

$$V(\mathbf{x}) = V^{\Omega^+} H(s(\mathbf{x})) + V^{\Omega^-} H(1 - s(\mathbf{x})) \quad (12)$$

Note that

$$\nabla H(s(\mathbf{x})) = \delta(s)\nabla s = \delta(s)\hat{\mathbf{n}}$$

So the gradient of the potential can be written as

$$\nabla V(\mathbf{x}) = H\nabla V^{\Omega^+} + (1 - H)\nabla V^{\Omega^-} + (V^{\Omega^+} - V^{\Omega^-})\delta(s)\hat{\mathbf{n}} \quad (13)$$

(13) implies that $V^{\Omega^+} - V^{\Omega^-} = 0$. This has two meanings. First, since the electric field $\mathbf{E} = -\nabla V$, the field will "explode" if and only if $V^{\Omega^+} - V^{\Omega^-} \neq 0$; second, the terms V^{Ω^+} and V^{Ω^-} have definition only at the trace($V^{\Omega^\pm} \rightarrow V^\pm$) under the meaning of the distribution $\delta(s)$, so it leads to the same conclusion as (11). We further take the divergence of the gradient in (13), namely, the Laplace, then we have

$$\begin{aligned} \nabla^2 V(\mathbf{x}) &= H\nabla^2 V^{\Omega^+} + (\nabla H) \cdot (\nabla V^{\Omega^+}) + \nabla^2 V^{\Omega^-} - \nabla \cdot (H\nabla V^{\Omega^-}) \\ &= H\nabla^2 V^{\Omega^+} + (1 - H)\nabla^2 V^{\Omega^-} + (\nabla V^{\Omega^+} - \nabla V^{\Omega^-})\delta(s)\hat{\mathbf{n}} \\ &= H\nabla^2 V^{\Omega^+} + (1 - H)\nabla^2 V^{\Omega^-} + (\partial_{\hat{\mathbf{n}}} V^{\Omega^+} - \partial_{\hat{\mathbf{n}}} V^{\Omega^-})\delta(s) \end{aligned}$$

By Poisson's Equation and the proposition 3.1, we have

$$\nabla^2 V = -\frac{\varrho(\mathbf{x})}{\epsilon_0} = -\frac{\sigma(\mathbf{x})}{\epsilon_0} \delta(s(\mathbf{x})) \quad (14)$$

By comparing the coefficients, we can obtain a similar result to (7), which is

$$\partial_{\mathbf{n}} V^+ - \partial_{\mathbf{n}} V^- := \partial_{\mathbf{n}} V^\Omega = -\frac{\sigma(\mathbf{x})}{\epsilon_0} \quad (15)$$

Therefore, the boundary condition can be written as

$$\boxed{-\epsilon_0 \partial_{\mathbf{n}} V^\Omega = \sigma(\mathbf{x})} \quad (16)$$

under the meaning of distribution. Since we are going to discuss the conductor, the field inside it is 0. So the boundary conditions give that

$$\begin{cases} \mathbf{E} = \frac{\sigma(\mathbf{x})}{\epsilon_0} \hat{\mathbf{n}} \Big|_{\text{outward}} \\ -\epsilon_0 \partial_{\mathbf{n}} V^\Omega \Big|_{\text{out}} = \sigma(\mathbf{x}) \end{cases} \quad (17)$$

A charge distribution located outside the conductor affects the surface, resulting in a force per unit area $\mathbf{f}$ and an external electric field $\mathbf{E}_{\text{other}}$ acting on the conductor. In sections 1 & 2, we just consider the charge on the surface itself, so we can choose a small patch of the surface, which is $\mathbf{E} = \pm \sigma / 2\epsilon_0 \hat{\mathbf{n}}$. And then

$$\begin{cases} \mathbf{E}^+ = \mathbf{E}_{\text{other}} + \frac{\sigma(\mathbf{x})}{2\epsilon_0} \hat{\mathbf{n}} \\ \mathbf{E}^- = \mathbf{E}_{\text{other}} - \frac{\sigma(\mathbf{x})}{2\epsilon_0} \hat{\mathbf{n}} \end{cases} \quad (18)$$

The field and the force(per unit area) that act on the conductor are

$$\mathbf{E}_{\text{other}} = \frac{1}{2} [\mathbf{E}(\mathbf{x})^+ + \mathbf{E}(\mathbf{x})^-]; \quad \mathbf{f} = \frac{1}{2} \sigma(\mathbf{x}) [\mathbf{E}(\mathbf{x})^+ + \mathbf{E}(\mathbf{x})^-] \quad (19)$$

By condition (17), the force acting on the conductor is determined, that is,

$$\boxed{\mathbf{f}(\mathbf{x}) = \frac{\sigma(\mathbf{x})^2}{2\epsilon_0} \hat{\mathbf{n}}} \quad (20)$$

We can further obtain the **electrostatic pressure** P

$$\boxed{P = \mathbf{f}(\mathbf{x}) \cdot \hat{\mathbf{n}} = \frac{\sigma(\mathbf{x})^2}{2\epsilon_0} = \frac{\epsilon_0}{2} |\mathbf{E}^+|^2} \quad (21)$$